Charan Surya Nikhil Kilana

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GitHub | Portfolio
Location: Dallas, Texas

Professional Summary:

- Having 10+ Years's experience in designing and deploying applications in AWS (**EC2, IAM, EBS, S3, VPC, Route53, CloudWatch, RDS, DynamoDB, Cloud Formation, ECR, ECS, EKS, SQS, SNS**) and Azure (**VMs, AD, Blob, VNet, Azure DNS, Monitor, Azure Database, Cosmos DB, ARM, ACR, ACS, SKS, Queue, Event Grid**) stack.
- Hands-on experience in building CI/CD pipelines using **Jenkins**, **GitHub Actions**, and **GitLab CI** to automate build, test, security scanning, and deployment processes.
- Extensive experience in building pipeline templates in **Azure DevOps** across Java, Dotnet, Python, Liquibase, Helm.
- Experience in developing **repo templates** for critical pipeline template so that developers can onboard repo without any effort.
- Standardized pipelines to ensure consistent validation and seamless deployment of backend services integrated with **LangChain** and **OpenAI**.
- Enabled efficient deployment of AI knowledge retrieval services(**MCP, RAG**), ensuring up-to-date and accurate responses for enterprise chat agent use cases.
- Extensive experience in installing, configuring, and administering **Jenkins** CI tool on **Linux** machines. Used **Jenkins** Pipeline to drive all **Microservices** builds out to the **Docker** registry and then deployed to **Kubernetes**, Created **Pods** and managed using **Kubernetes,** and used **Ansible playbooks** to create **clusters in Kubernetes by using Jenkins.**
- Experienced in **Ansible Tower**, developing **Ansible playbooks** for managing the application/OS configuration files in GitHub, integrating with **Jenkins**, and Verifying with **Jenkins** plugins, deploying the application in Linux environment.
- Expert in deploying the code through web application servers like **Web Sphere/Web Logic/ Apache Tomcat/JBoss** and their installation, configuration, management and troubleshooting.
- Experienced in setting up various **Jenkins jobs** like **CRON jobs**, etc. to pull the code from the **GitHub repository** and the continues integration and continues delivery (CI/CD) pipeline is built by writing a Jenkins file using **Groovy** scripting language.
- Expertise in using **Docker** including **Docker Hub**, **Docker Engine**, **Docker images**, **Docker compose**, **Docker swarm**, and **Docker Registr**y and used containerization to make our applications platform to be consistent flexible when they are moved into different environments.
- Designed, deployed, and managed scalable, containerized applications using **Kubernetes** (**EKS**, AKS, KOPS and **self-managed clusters**), implementing **CI/CD** pipelines, auto-scaling, rolling updates, and monitoring.
- Proficient in Scripting languages like **Bash, Groovy, Python and Power shell, JSON** and **XML** for automating administrative tasks.
- Expertise in configuring the monitoring and alerting tools according to the requirement like **Prometheus** and **Grafana**, setting up alerts and deploying multiple dashboards for individual applications in **Kubernetes**.
- Experience in using different log monitoring tools like **Nagios, CloudWatch, Splunk, ELK** (**Elastic Search, Log Stash, Kibana**) to see logs information, monitor, security and get health notifications from nodes. Automate the installation of **ELK** agent (filebeat) with Ansible playbook.

- Experienced in **Bluegreen Deployment and Canary Deployment, Octopus deployment** in the production. Managed Kubernetes cluster in the rollback and rollouts methods in the deployment strategies. Integrated **Istio, ArgoCd** and **helm** packages with Kubernetes clusters for the service mesh.
- Collaborated with **Okta** team to implement **Single Sign-On (SSO)** for **Terraform Cloud** and **AWS** accounts, enhancing security and streamlining access management.

Technical Skills:

AWS	EC2, IAM, EBS, S3, VPC, Route53, CloudWatch, RDS, DynamoDB, Cloud Formation, ECR, ECS, EKS, SQS, SNS
Azure	VMs, AD, Blob, VNet, Azure DNS, Monitor, Azure Database, Cosmos DB, ARM, ACR, ACS, AKS, Queue, Event Grid
Source Code Management	Azure Repos, GitHub, GitLab, Bitbucket
Build Tool	Maven, Ant, Gradle, nuget
CI/CD	Azure DevOps, Jenkins, GitHub Actions, GitLab
Artifactory Management	Nexus, JFrog
Configuration Management/ Infrastructure as Code	Ansible, Chef, Terraform, Cloud Formation, Azure Resource Manager
Containerization & Orchestration	Docker, Kubernetes
Programming/ Scripting Languages	Java, Python, PowerShell, Bash, Groovy, YAML, XML
Monitoring Tool	Prometheus, Grafana, EFK, ELK, Dynatrace, CloudWatch, Azure Monitor, Splunk
Operating System	Amazon Linux, RHEL, Ubuntu, Windows, PowerShell
AI tools	RAG pipelines, MCP, LangChain, LangSmith

Professional Summary:

Client: Humana, Louisville, Kentucky, USA **October 2023 – Present**
Role: Azure Platform Engineer

Project Description:
Working in a centralized **E3 CI/CD DevOps** team to design and maintain scalable pipeline templates for onboarding applications across the organization. Enable standardized repository provisioning and **CI/CD** frameworks for multiple technology stacks. Develop and validate pipelines using **PowerShell** and **Pester** to ensure reliability and consistency. Utilize a **.NET Core API** to automate repository provisioning and integrate pipeline templates. Support deployments across platforms like **ArgoCD, cloudrun, AzureApp Service** and **Terraform**, ensuring secure, scalable, and efficient delivery.

Responsibilities:

- Engineered scalable pipeline templates in Azure DevOps supporting multiple tech stacks including **Python, Java, and .NET**, integrated with deployment targets such as **Argo CD**, **Google Cloud Run**, and **Azure App Services**.
- Maintained and enhanced cross-platform CI/CD templates (**Artifactory, ArgoCD, App Service, Liquibase pipelines**), ensuring consistency and reusability across teams.
- Developed **repository provisioning** frameworks to bootstrap new projects with standardized pipeline templates in both Azure DevOps and GitHub environments.

- Contributed to onboarding strategy by enabling developers to adopt secure, pre-configured CI/CD templates, reducing setup time and security misconfigurations.
- Built **CI/CD** pipeline templates to support internal AI chat agent (**Strider**), enabling automated deployment of **MCP server components** integrated with **LangChain and OpenAI.**
- **Supported RAG-based** automation initiatives by creating reusable pipeline templates for deploying and validating documentation/runbook-driven AI services across enterprise environments.
- Automated cleanup of stale and unused pipelines across multiple projects by implementing PowerShell scripts targeting **.fdsecure** pipeline structures, improving maintainability.
- Integrated backend provisioning workflows using **.NET Core APIs** to dynamically create repositories and inject pipeline templates.
- Utilized **Splunk** for centralized log analysis and performance monitoring, enabling efficient debugging of repository provisioning workflows using App Service identifiers.
- Migrated secrets management from traditional service connections to **HashiCorp Vault**, enhancing security and compliance across CI/CD pipelines.
- Implemented hybrid support for both **legacy service** connections and **Vault-based** secret management to ensure backward compatibility during transition.
- Designed and enhanced Helm-based deployment templates (**Helm-to-AKS / Helm-to-GKE**) to standardize Kubernetes application deployments, incorporating secure secrets management, configuration handling, and environment-specific customization.
- Designed and developed automated CI/CD pipelines using **PowerShell** to enable bulk migration of repositories, configurations, and pipeline assets.
- Built reusable **PowerShell**-based automation to manage and deploy **Azure DevOps** extensions across multiple organizations via centralized pipeline execution.
- Currently developing **Terraform-based pipeline templates** to enable infrastructure-as-code driven onboarding of CI/CD pipelines, while enhancing **E3 repository** provisioning frameworks to standardize and accelerate project setup.

Client: NextGen Health Care Atlanta, GA **March 2021 – October 2023**
Role: DevOps Engineer

Project Description:
I worked at **NextGen Healthcare**, a leading provider of healthcare technology. Led the automation and deployment of cloud native healthcare applications using both **AWS** cloud. The platform was designed to handle sensitive patient data securely, ensuring **HIPAA** compliance. Implemented **CI/CD** pipelines, **Infrastructure as Code**, and **container orchestration** to streamline releases ultimately enhancing the efficiency of clinical workflows and patient care delivery.
Responsibilities:

- Designed and implemented **CI/CD pipelines (Jenkins, GitHub Actions)** for automating ArcGIS map service deployments, configuration synchronization, and infrastructure versioning.
- Managed **Terraform modules** for reusable AWS resource provisioning **VPCs, subnets, Route53, S3, EKS** clusters supporting both ArcGIS workloads and adjacent microservices.
- Implemented **Blue-Green** and **Canary** deployment strategies for EKS-hosted microservices to ensure zero-downtime updates for internal mapping and analytics APIs.
- Developed **Python** & **Bash** automation scripts to sync millions of transactional records from corporate PostgreSQL to ArcGIS Data Store, enabling daily updates to GIS dashboards.
- Configured **Prometheus** and **Grafana** to monitor **ArcGIS Server**, Portal, and EC2 metrics; integrated Splunk for centralized log management and real-time alerting.

- Collaborated with Okta **IAM** and Network Security teams to enforce SSO, MFA, and role-based access policies across ArcGIS Portal, Terraform Cloud, and AWS accounts.
- Managed **Red Hat Enterprise Linux (RHEL)** servers hosting ArcGIS stack, ensuring compliance, patching, and system hardening per corporate security baselines.
- Enabled cost optimization through **Terraform** state backends **(S3 + DynamoDB), EC2 right-sizing**, and container auto-scaling within EKS, resulting in ~30% infrastructure cost reduction.
- Used **Splunk** for log analysing and improving the performance of servers. Developed several custom **Splunk** queries for monitoring and alerting and written **Python** scripts to automate security group creation and management from state files versioned **GitHub.**
- Configured and maintained **Jenkins** to implement the **Continuous Integration** process, integrated the tool with **Gradle, Maven** to schedule the build jobs and automated the deployment on the application servers using the code deploy plugin for **Jenkins** and used **Jira** as ticketing tool.
- Developed **Ansible** scripts for **EC2** instances, **Elastic Load balancers** and **S3 buckets,** Planned, developed & maintained Infrastructure as code using **CI/CD** deployments using **Ansible**.

Client: Staples - Framingham, MA **January 2018 – February 2021**
Role: DevOps Engineer

Project Description:
At Staples, a global retail company, on their e-commerce and supply chain management platform. Led the infrastructure and deployment of cloud-native retail applications using **AWS** and **DevOps** practices. The platform supported large-scale order processing, inventory tracking, and customer engagement, ensuring high availability during seasonal traffic spikes. Implemented **CI/CD** pipelines, **Infrastructure as Code**, and monitoring solutions to reduce downtime, enhance the overall customer shopping experience.
Responsibilities:

- Launched **AWS EC2** Cloud Instances using **Amazon Web Services (Linux/ Ubuntu/RHEL)** and Configured the launched instances with respect to specific applications. Created **Snapshots** and **Amazon Machine Images (AMI's)** for mission-critical production servers for backup.
- Maintained **DNS** records using **Route53**. Used **AWS** Route53 to manage **DNS zones** and give public **DNS** names to **elastic load balancers** IPs.
- Responsible for delivering an end-to-end continuous integration of a continuous delivery system for the products in an **agile** development approach using **Puppet** and **Jenkins**.
- Implemented a Continuous Delivery pipeline with **Docker**, **Jenkins** and **GitHub**. Whenever there is a change in **GITHUB**, our Continuous Integration server automatically attempts to build a new **Docker** container from it.
- Responsible for installation and configuration of **Jenkins** to support various **Java builds** and **Jenkins** plugins to automate continuous builds and publish **Docker Images** to the **Nexus Repository**.
- Implemented **Docker maven-plugin** in **Maven POM** to build **Docker Images** for all **microservices** and later used **Docker** file to build the **Docker Images** from the **java jar files.**
- Set up **Jenkins server** and build jobs to provide Continuous Automated builds based on Polling the **Git** source control system during the day and periodic scheduled builds overnight to support development needs using **Jenkins**, **Git**, and **Maven**.
- Worked on **NoSQL** database **MongoDB** to replica setup and sharding. Also experienced in managing replica set. Installed, Configured, and Managed **Monitoring Tools** such as **Nagios and Splunk** for Resource/Network Monitoring.
- Used **Selenium** for continuous inspection of code quality and to perform automatic reviews of code to detect bugs. Automated **Nagios** alerts and email notifications using **Python script** and executed them through **Chef**.
- Configured dashboards **RDS** in **Elastic, Log stash & Kibana (ELK)**. Used **ELK** to set up real time logging and analytics for Continuous delivery pipelines and applications.

Client: Xpressbees, Pune IN **July 2016 – December 2018**
Role: Azure DevOps Engineer

Project Description:
At Xpressbees, a leading logistics and express delivery company in Pune, India. In this project, Led the automation of **cloud infrastructure** and **deployment** of highly available applications using **AWS**. The system handled real-time package tracking, route optimization, and delivery notifications. Implemented Infrastructure provisioning using Terraform and deployed microservices through **Kubernetes** to scale operations.
Responsibilities:

- Deployed and managed **Azure** cloud infrastructure, leveraging services such as **Azure AD, VMs, Blob, VNet, Load Balancer & Autoscaling groups** using **Terraform** with **Azure** based on Business needs.
- Designed, implemented, and maintained **Azure DevOps** pipelines, enabling efficient and automated code compilation, testing, and deployment processes.
- Developed **Dockerfiles** based on the code and built the **multistage docker files** to create custom images and scanned the image by using the tool called **Trivy**.
- Configured Kubernetes resources including **Namespaces, Pods, Deployments, Services, and Ingress, PV & PVC** implementing best practices for container orchestration.
- Troubleshoot and resolve **Kubernetes** related issues, including pod and node-level problems, ensuring minimal downtime and maintaining application reliability.
- Implemented monitoring solutions using **Prometheus** and **Grafana**, tracking system performance, collecting metrics, and visualising data for effective decision-making.
- Automated infrastructure provisioning using **Terraform**, ensuring consistent, repeatable, and scalable deployments across multiple environments.
- Automated repetitive tasks and manual interventions using **scripting languages** and **configuration management tools**, enhancing operational efficiency and reducing human error.

Education:

Bachelor of Technology in Computer Science **2016**
GITAM UNIVERSITY

Certifications:

HashiCorp Certified Terraform Associate | AWS Certified Solution Architect | AWS Certified Cloud Practitioner | Microsoft Certified: Azure Administrator Associate